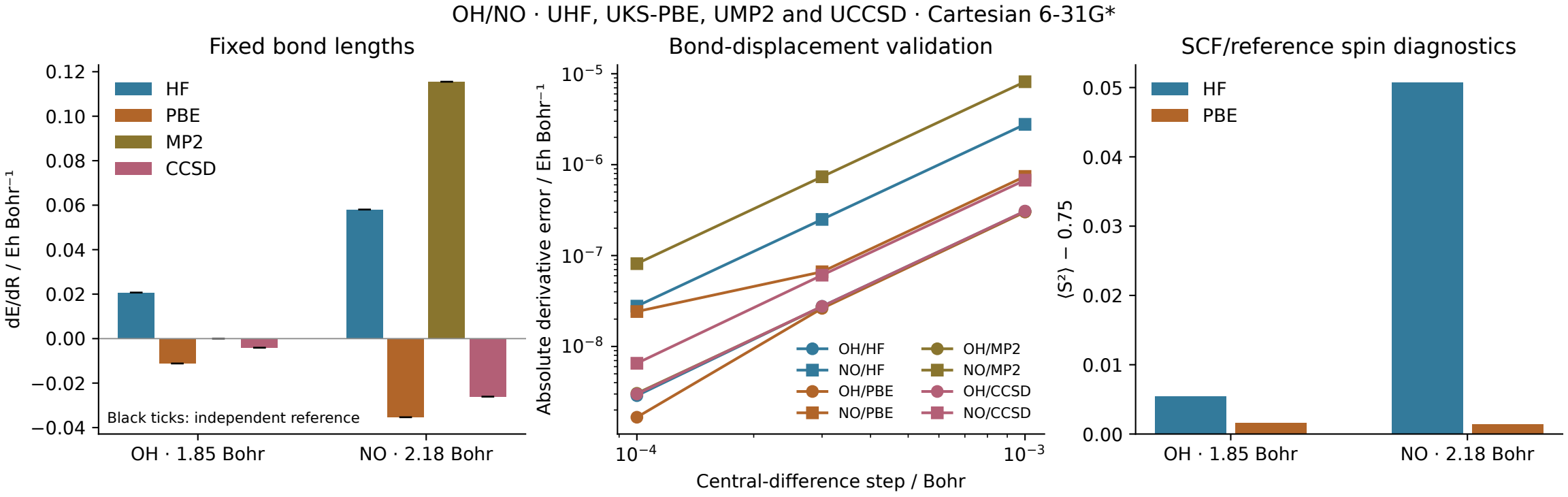
OH/NO · UHF, UKS-PBE, UMP2 and UCCSD · Cartesian 6-31G*
Fixed bond lengths
dE/dR / Eh Bohr⁻¹
HF
PBE
MP2
CCSD
Black ticks: independent reference
OH · 1.85 Bohr
NO · 2.18 Bohr
Bond-displacement validation
Absolute derivative error / Eh Bohr⁻¹
OH/HF
NO/HF
OH/PBE
NO/PBE
OH/MP2
NO/MP2
OH/CCSD
NO/CCSD
Central-difference step / Bohr
SCF/reference spin diagnostics
⟨S²⟩ − 0.75
HF
PBE
OH · 1.85 Bohr
NO · 2.18 Bohr